Review

1. Induction as a proof method (Usual and strong)
2. Row operations for solving linear systems
3. Fields & their basic properties
4. $\mathbb{C}$: Euler's notation, conjugates, etc.
5. Vector spaces over fields
6. Vector subspaces
7. Linear combinations & spans
8. Linear independence

(9) Bases & dimension

(10) Steinitz exchange lemma & its applications

Functions

$$f : \underset{\text{domain}}{X} \longrightarrow \underset{\text{codomain}}{Y}$$

is a rule assigning to each

$$f : x \longmapsto y = f(x)$$

$x \in X$, \mapsto, $y \in Y$

Examples:

(1) $f : \mathbb{R} \longrightarrow \mathbb{R}$

$a \longmapsto a^2$

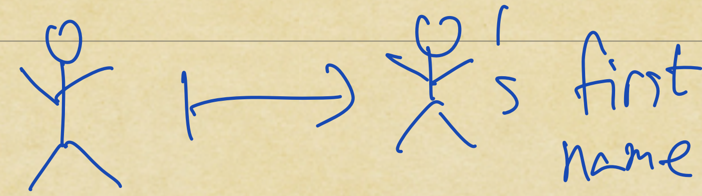
$\boxed{f(x) = x^2}$

NOT ONTO

NOT ONE-TO-ONE

(2)

$$\begin{array}{ccc} & \{\text{Human beings (with first names)}\} & & \{\text{words}\} \\ & \parallel & & \parallel \\ f: & X & \longrightarrow & Y \\ & \in & & \in \end{array}$$



NOT ONTO

NOT ONE-TO-ONE

(3)

$$\begin{array}{ccc} & \{\text{Collections of words}\} & & \mathbb{N} \\ & \parallel & & \parallel \\ f: & X & \longrightarrow & Y \\ & \in & & \\ & \{\text{mush, gravy, monkey}\} & \longmapsto & 4 \end{array}$$

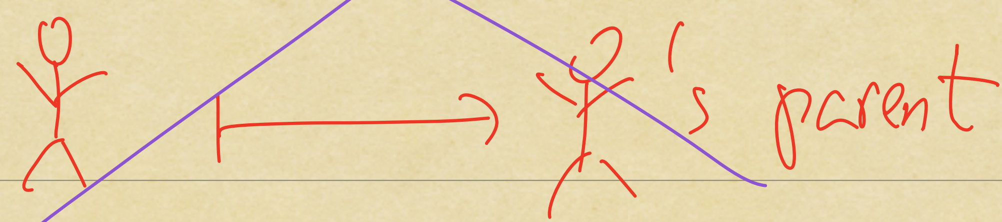
Probably NOT ONTO

NOT ONE-TO-ONE

$f:$ collection $\longmapsto$ length of the smallest word in the collection

Rmk: Every input should have a unique output.

e.g. $\{\text{Human beings}\} \longrightarrow \{\text{Human beings}\}$



NOT A FUNCTION

Defn $f: X \longrightarrow Y$

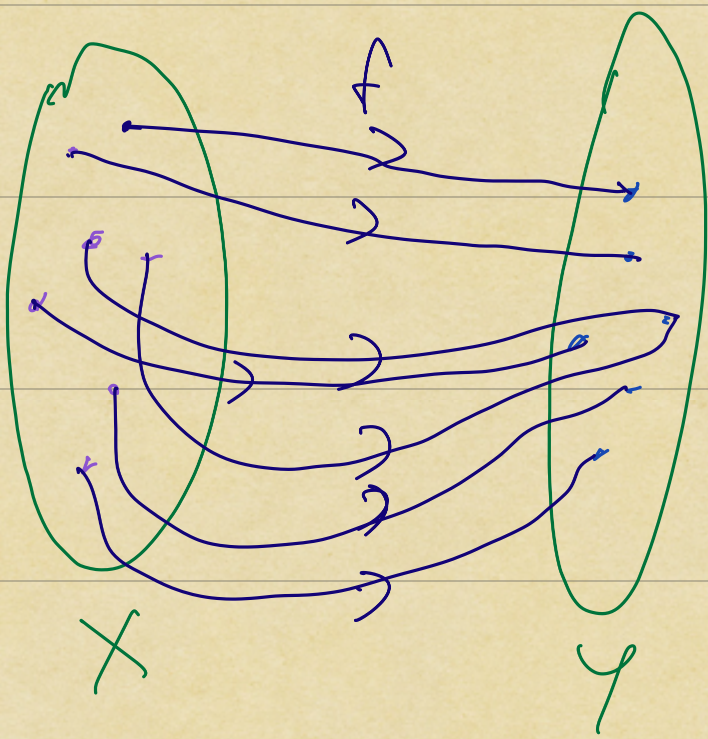
① f is onto or surjective if $\forall y \in Y, \exists x \in X$ s.t.

$f(x) = y$

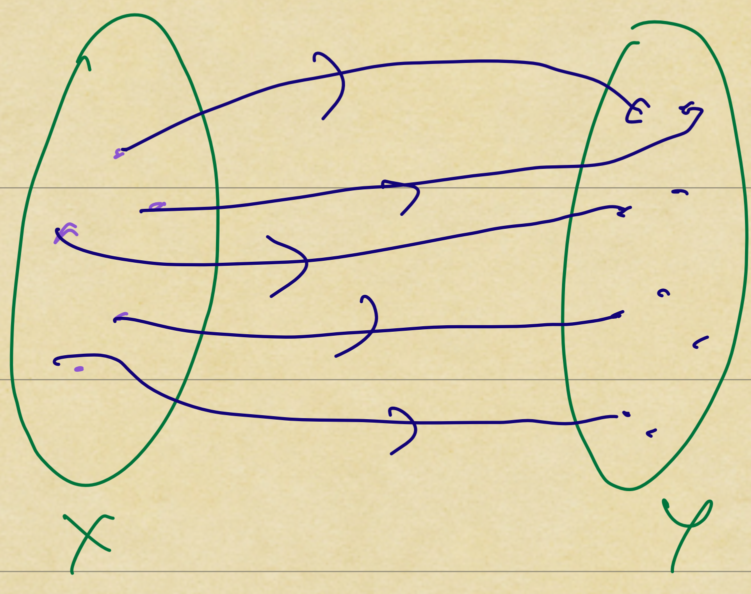
(2) f is one-to-one or injective

if $\forall x_1, x_2 \in X$

$f(x_1) = f(x_2) \implies x_1 = x_2$



ONTO



ONE-TO-ONE

Example (1) (again)

$$\left.\begin{array}{l} f : \mathbb{R} \longrightarrow \mathbb{R} \\ a \longmapsto a^2 \end{array}\right\} \text{NOT } \underline{\text{ONTO}}$$

We could also consider

$\underline{\text{ONTO}}$

NOT ONE-TO-ONE

$$\left[\begin{array}{l} g : \mathbb{R} \longrightarrow \mathbb{R}_{\geq 0} = \{b \in \mathbb{R} : b \geq 0\} \\ a \longmapsto a^2 \end{array}\right.$$

ONE-TO-ONE

$\underline{\text{NOT}}$ ONTO

$$\left[\begin{array}{l} h : \mathbb{R}_{\geq 0} \longrightarrow \mathbb{R} \\ a \longmapsto a^2 \end{array}\right.$$

ONE-TO-ONE & ONTO

$$\left[\begin{array}{l} k : \mathbb{R}_{\geq 0} \longrightarrow \mathbb{R}_{\geq 0} \\ a \longmapsto a^2 \end{array}\right.$$

③ $f: X \to Y$ is **bijective** or **a bijection**

if it is both **one-to-one** & **onto**

Example: X: set

$$\begin{array}{ccc} \mathrm{id}_X : X & \longrightarrow & X \\ \cup\!\!\!\!\!\in \quad x & \longmapsto & x \end{array}$$

identity function of X

match

Def$^{\underline{n}}$ $f: X \to Y$, $g: Y \to Z$

$$\left(\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "g \circ f"'] & Y \arrow[d, "g"] \\ & Z \end{tikzcd}\right)$$

The **composition** $\boxed{g \circ f}: X \to Z$ is given by

$$x \overset{f}{\longmapsto} f(x) \overset{g}{\longmapsto} g(f(x))$$

Rmk: Note the order of the functions!

Fact: $f : X \to Y$

TFAE: (a) f is a bijection

(b) $\exists\, g : Y \to X$ with the following property:

$\forall x \in X,\ y \in Y$

$$\begin{tikzcd} x \arrow[r, mapsto, "f"] & f(x) \arrow[d, mapsto, "g"] \\ & g(f(x)) = x \arrow[ul] \end{tikzcd} \qquad \begin{tikzcd} y \arrow[r, mapsto, "g"] & g(y) \arrow[d, mapsto, "f"] \\ & f(g(y)) = y \arrow[ul] \end{tikzcd}$$

i.e.

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "\mathrm{id}_X = g \circ f"'] & Y \arrow[d, "g"] \\ & X \end{tikzcd} \quad \& \quad \begin{tikzcd} Y \arrow[r, "g"] \arrow[dr, "\mathrm{id}_Y = f \circ g"'] & X \arrow[d, "f"] \\ & Y \end{tikzcd}$$

Pf of Fact (b) $\Rightarrow$ (a)

(ONTO) $y \in Y$; then $y = f(g(y))$

(ONE-TO-ONE) $x_1, x_2 \in X$ s.t. $f(x_1) = f(x_2)$

$$\begin{array}{ccc} f(x_1) & = & f(x_2) \\ \downarrow g & & \downarrow g \\ g(f(x_1)) & = & g(f(x_2)) \end{array}$$

$$\begin{matrix} \| & & \| \\ x_1 & \overset{\checkmark}{=} & x_2 \end{matrix}$$

(c) $\Rightarrow$ (b)

f is one-to-one & onto

Define $g: Y \longrightarrow X$ by the rule:

$g(y) =$ the unique x s.t.

$f(x) = y$

Then we have $\forall y \in Y$

$$f(g(y)) = y$$ BY DEFINITION!

But what is $g(f(x))$?

(for $x \in X$)

For this, note:

$$f(g(f(x))) = f(x) \xRightarrow[\text{is one-to-one}]{f} g(f(x)) = x$$

(why?)